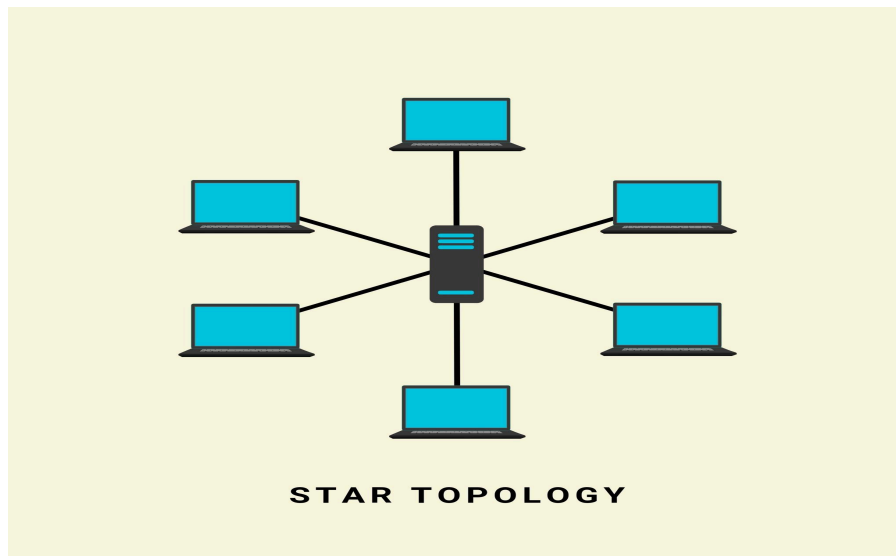


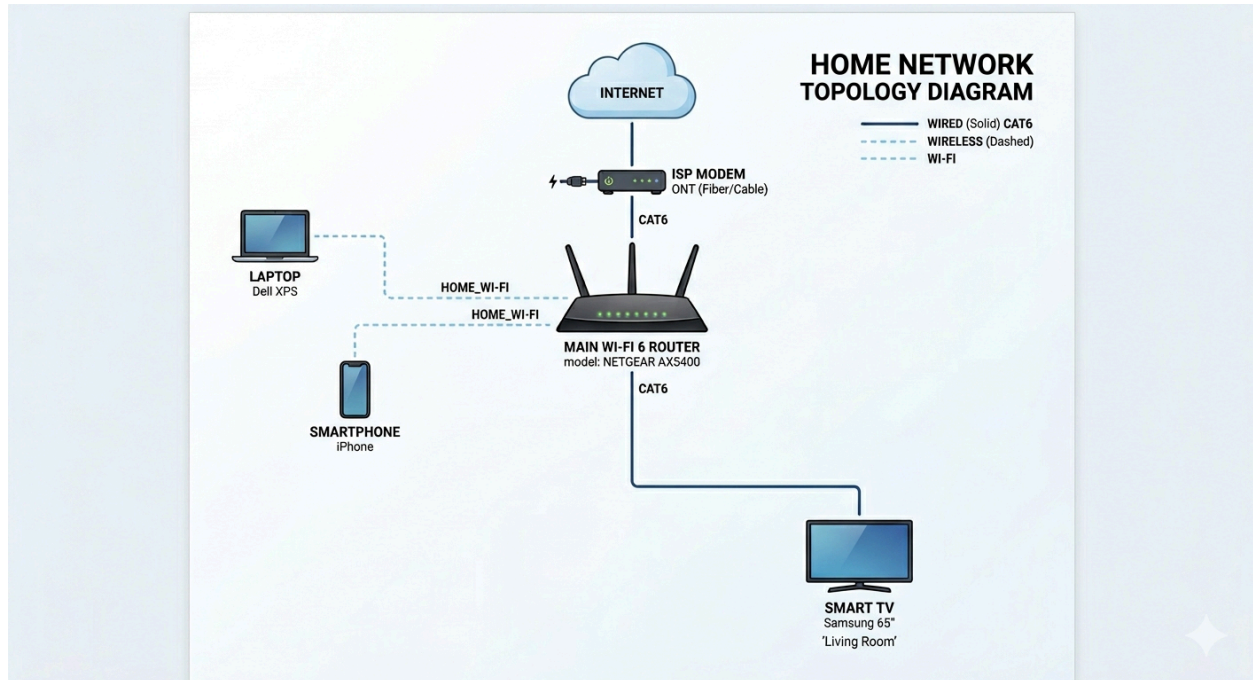
Name - Saikat Sheet  
Batch - 25SUB5936\_Linux System  
Programming  
ID - 68500

## **Home Network Topology**

### **STAR TOPOLOGY**

A Star Topology is a network layout where every device connects to one central hub, like a router. All data must pass through this hub to reach its destination. If a single cable fails, only that device disconnects; however, if the central hub fails, the entire network goes down.





## 1. The Central Hub: Wi-Fi Router

The **Wi-Fi Router** is the most important part of our home network. It performs two main jobs:

- **The Gateway:** It takes the internet signal coming from our service provider and shares it with our home.
- **The Director:** It assigns a unique "address" to every device so that when we search for something on our laptop, the results don't accidentally pop up on our TV.

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## 2. The Wired Device: Smart TV

In this network, the **Smart TV** is connected to the router using a physical **Ethernet Cable** (often called a CAT6 cable).

- **Why use a cable?** While TVs can use Wi-Fi, a wired connection is much faster and more reliable. It prevents the video from "buffering" or lagging when we are watching high-definition movies or streaming 4K content.
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### 3. The Wireless Devices: Laptop and Mobile Phones

These devices connect to the router using **Wi-Fi waves** instead of physical cables. This allows for "mobility," meaning we can move from room to room without losing our connection.

- **Laptop:** The laptop uses Wi-Fi so we can work at a desk, on the couch, or in the kitchen. It handles tasks like emails, video calls, and web browsing.
- **Mobile Phones:** Multiple smartphones connect to the same Wi-Fi signal. This allows the phones to download large app updates and stream music without using up your expensive cellular data plan.